

Commodity Catalyst and Volatility Monitor

Detailed Product Proposal - Data-First MVP

Initial market: NYMEX WTI crude oil futures and options | Context: Brent futures and public energy catalysts

Product thesis. A daily, forward-looking research workflow that interprets the WTI futures curve and WTI option volatility surface as market-implied expectations, maps upcoming catalysts to exposed delivery windows and option expiries, and produces a concise risk brief with scenarios, confirmation triggers, and data-quality caveats.

1. Executive Summary

The Commodity Catalyst and Volatility Monitor (CCVM) is an end-of-day decision-support system for commodity trading desks, volatility teams, and hedge funds. Its primary output is not a retrospective news summary. It identifies where risk is currently priced, which upcoming catalysts overlap those exposures, how futures and options may reprice under alternative scenarios, and what evidence should confirm or invalidate each thesis.

- Primary comparison: WTI futures curve versus the WTI option volatility surface.
- Contextual cross-check: Brent futures, inventory and refinery data, positioning, and verified catalyst information.
- Operating frequency: daily after settlement, with optional event-time refreshes later.
- Core design principle: deterministic calculations produce market signals; the agent retrieves, ranks, and synthesizes evidence.
- MVP priority: reliable data collection, provenance, normalization, and validation before sophisticated modeling.

2. Target Users and Decisions Supported

User	Daily question	CCVM contribution
Commodity PM / trader	Where is risk concentrated and what could move it next?	Delivery-window risk map, catalyst priorities, scenarios, monitoring triggers.
Volatility trader	Is option premium directional, event-driven, or broad uncertainty?	ATM term structure, risk reversals, butterflies, event-window comparisons.
Desk strategist / quant	Do futures and options tell a coherent story?	Standardized signal objects, agreement taxonomy, quality controls.
Risk manager	Which exposures are vulnerable to the next catalyst?	Scenario shocks, contract and expiry mapping, evidence-backed escalation.

3. Product Scope

3.1 MVP scope

- WTI futures: first 8-12 monthly contracts, daily settlement, volume, open interest, contract metadata.
- WTI options: first 3-5 liquid expiries, strike-level settlement, volume, open interest, underlying-future mapping.
- Brent futures: first 6-8 contracts as a global-versus-US diagnostic control.
- Catalysts: EIA petroleum releases, OPEC-related events, outages, sanctions deadlines, refinery disruptions, weather/hurricane risks, scheduled macro-demand releases, and high-quality news.
- Output: daily HTML/Markdown/PDF-style forward risk brief and a Streamlit review interface.

3.2 Explicit non-goals for MVP

- No autonomous order placement or trade recommendation.
- No claim of executable mispricing from settlement-only data.
- No historical percentile or mean-reversion claim until a sufficient clean history is collected.
- No multi-commodity generalization until WTI data quality and mappings are stable.

4. Forward-Looking Analytical Framework

The daily process is organized around four questions:

1. Where is the market pricing risk?
2. Which upcoming events overlap that delivery or expiry window?
3. What scenarios could reprice futures, spreads, volatility, and skew?
4. What observations would confirm, invalidate, or escalate the thesis?

4.1 Futures risk map

- Contract-level daily and five-day changes.
- Adjacent calendar-spread changes and front-versus-back slope changes.
- Local butterflies and delivery-month concentration.
- Prompt tightness versus deferred balance; WTI versus Brent context.

4.2 Option risk map

- Black-76 implied volatility using the correct underlying futures contract and option expiry.
- ATM implied volatility term structure by expiry and delivery month.
- 25-delta call-put risk reversal and 25-delta butterfly, with explicit sign conventions.
- Event-window comparisons between neighboring expiries and surface quality flags.

4.3 Futures-options agreement taxonomy

confirmed_upside_risk, confirmed_downside_or_demand_risk, non_directional_uncertainty, futures_only_repricing, options_only_repricing, cross_market_disagreement, no_material_change, insufficient_data

4.4 Forward scenario layer

- Editable bull/base/bear/event scenarios rather than a single forecast.
- Curve shocks can be front-loaded, parallel, or delivery-month specific.
- Volatility shocks can alter ATM, risk reversal, butterfly, and term structure independently.
- Each scenario includes expected cross-market behavior, monitoring conditions, and confidence.

5. Data-First Strategy

Data collection is the critical path. The system should be useful with settlement snapshots, but it must preserve raw records and provenance so that a historical research asset accumulates from day one.

5.1 Data-source tiers

Tier	Purpose	Examples	Policy
Tier 0 - manual/sample	Develop and test schemas immediately	Curated CSV snapshots and synthetic fixtures	Never mixed with production records without a source flag.
Tier 1 - public bootstrap	Build a daily prototype at low cost	CME public futures and option settlement pages/tools; EIA Open Data and WPSR; CFTC COT	Store retrieval timestamp, page/source identifier, and raw payload. Respect terms and avoid brittle scraping as the long-term design.
Tier 2 - licensed EOD	Reliable recurring production collection	CME DataMine settlement datasets or another licensed distributor	Preferred production path for historical and daily files; entitlement-aware ingestion.
Tier 3 - live/intraday	Event-time monitoring and executability	CME real-time WebSocket or licensed feed	Deferred until the EOD product proves useful.

5.2 Public and official bootstrap sources

- CME publishes WTI futures settlement pages and WTI option settlement pages/tools. Public delayed quotes are delayed by at least 10 minutes. These are useful for prototype access and manual verification, but production ingestion should use a licensed or explicitly supported data delivery channel [1][2][3].
- CME DataMine offers historical futures and options datasets, including settlements, and provides APIs for entitled files. This is the natural upgrade path if the project moves beyond a demo [4][5].
- EIA Open Data provides a free API for petroleum time series, while the Weekly Petroleum Status Report supplies inventories, refinery utilization, imports/exports, and related weekly balances [6][7].
- CFTC Commitments of Traders data provides weekly aggregate futures and options positioning context. It is not strike-level options data and should be treated as a slow-moving contextual feature [8].

5.3 Canonical raw schemas

Every record must include source, retrieval timestamp, observation timestamp, market date, and a stable natural key. Raw files are immutable.

Futures settlement: trade_date, exchange, product, contract_code, delivery_month, settlement, volume, open_interest, source, retrieved_at

Option settlement: trade_date, option_expiry, underlying_contract, strike, call_put, settlement, volume, open_interest, exercise_style, source, retrieved_at

Catalyst event: event_id, event_type, title, published_at, effective_start, effective_end, commodity, region, direction, magnitude, source_url_or_id, source_quality

Fundamental observation: series_id, period, value, unit, release_timestamp, revision_vintage, source

5.4 Storage layout

raw/ -> bronze/ -> silver/ -> gold/ -> reports/

- raw: exact downloaded files or captured payloads, checksum, and metadata sidecar.
- bronze: parsed records retaining source-native identifiers.
- silver: normalized contracts, units, dates, option mappings, and quality flags.
- gold: daily curve features, volatility metrics, catalyst mappings, and report-ready objects.

- Recommended MVP storage: Parquet partitioned by trade_date plus DuckDB for query and audit.

5.5 Data quality gates

- Duplicate natural keys
- Missing or non-monotonic contract months
- Option mapped to wrong underlying future
- Invalid expiry or negative time to expiry
- Settlement outside intrinsic/no-arbitrage bounds
- Sparse strikes or unstable IV inversion
- Unexpected unit/currency change
- Large day-over-day move requiring source verification
- Late or missing daily file
- Revision-vintage mismatch for fundamental releases

6. System Architecture

**Collectors -> Immutable raw store -> Normalization/QA -> Quant analytics -> Catalyst agent
-> Scenario engine -> Daily brief**

6.1 Deterministic components

- Contract calendars, symbology, unit conversion, expiry mapping, and dataset reconciliation.
- Curve spreads, butterflies, changes, implied-volatility inversion, deltas, vegas, smile metrics, and scenario P&L.
- Agreement score inputs, source-quality weights, timestamps, and data-quality checks.

6.2 Agent components

- Choose which catalyst categories and sources to investigate for the observed risk window.
- Extract structured events and map them to contracts, expiries, regions, and expected direction.
- Rank supporting, contradicting, and contextual evidence.
- Construct a concise forward-looking narrative and monitoring checklist.
- Escalate unknown or conflicting situations rather than forcing a causal story.

7. Daily Output Specification

- Market-implied risk map
- Upcoming catalyst calendar
- Futures-options agreement assessment
- Three structured scenarios
- Ranked supporting and contradicting evidence
- Confirmation, invalidation, and escalation triggers
- Data-quality and executability caveats
- Next review time

Example summary

WTI prompt spreads and front-expiry call skew indicate concentrated upside risk in the next two delivery months. The next EIA release and a Gulf outage update are the highest-priority catalysts. Brent confirmation is limited, so the current thesis is US-specific tightness rather than a broad global shortage. A large inventory

build or a rapid retracement in call skew would invalidate the thesis. Settlement-only data does not establish executability.

8. MVP Delivery Plan

Phase	Focus	Deliverable	Exit criterion
0. Data contract	Schemas, source inventory, sample files	Versioned schemas, source registry, fixtures	Two dates of futures/options data parse cleanly.
1. Collection foundation	Raw acquisition, checksums, manifests, schedules	Idempotent collectors and immutable raw store	Repeated runs do not duplicate or corrupt data.
2. Normalization and QA	Symbology, contract calendars, mappings, validation	Silver Parquet tables and quality report	All critical mapping checks pass.
3. Quant analytics	Curve and surface metrics	Gold daily feature tables	Metrics reconcile to hand calculations.
4. Catalyst layer	Event extraction and relevance ranking	Structured events and ranked evidence	Top events are contract-window relevant.
5. Forward brief	Scenarios, triggers, UI/report	End-to-end daily report	Report is reproducible from stored inputs.

9. Risks and Mitigations

Data licensing and access: Use public sources only for bootstrap; preserve a provider interface; document the production upgrade to licensed EOD data.

Brittle web extraction: Prefer files/APIs/tools with explicit access; keep collectors isolated and test with recorded fixtures.

Sparse option settlements: Apply liquidity and strike-count gates; report `insufficient_data` rather than extrapolating aggressively.

Incorrect contract mapping: Treat option-to-future and expiry calendars as reference data with unit tests and manual reconciliation.

Post-hoc narrative bias: Require timestamps, supporting and contradicting evidence, confidence, and an unexplained classification.

False precision in scenarios: Make assumptions editable and clearly separate market-implied quantities from analyst scenarios.

10. Success Metrics

- Daily collection completion and latency
- Raw-to-silver reconciliation rate
- Contract and option-mapping error rate
- Percentage of reports with sufficient surface coverage
- Top-three catalyst relevance judged by a human reviewer
- Unsupported causal-claim count
- Percentage of reports with explicit contradiction or unexplained status
- User rating of whether the brief changed what they monitored next

11. Recommended Technology Stack

- Python 3.12, pandas or Polars, SciPy, Pydantic, DuckDB, PyArrow/Parquet.
- httpx and tenacity for collectors; structured logging and file checksums.
- Black-76 implementation with robust root finding; QuantLib optional, not required for the MVP.
- Streamlit for review; Jinja2/Markdown for reports; cron or GitHub Actions for a demo schedule.
- Small LLM for event extraction and ranking; stronger model only for final synthesis when evidence is conflicting.

12. Recommended First Build

The first useful milestone is not the agent. It is a trusted daily dataset containing WTI futures settlements, WTI option settlements, contract metadata, and EIA release observations, all with provenance and quality flags. Once ten to twenty clean daily snapshots exist, add curve/surface changes and the forward-risk brief. This sequence minimizes the risk of building polished reasoning on unreliable inputs.

References

- [1] CME Group, Crude Oil Futures Settlements.
- [2] CME Group, Crude Oil Option Settlements and Option Settlement Tools.
- [3] CME Group, Delayed Quotes - public quotes are delayed by at least 10 minutes.
- [4] CME Group DataMine - historical futures and options datasets, including settlements.
- [5] CME Group DataMine List API - access to entitled files.
- [6] U.S. Energy Information Administration, Open Data and APIv2 documentation.
- [7] U.S. Energy Information Administration, Weekly Petroleum Status Report.
- [8] U.S. Commodity Futures Trading Commission, Commitments of Traders reports.